Reneal eLearning Lab Operations

Report 2: Detailed Lab Usage Analytics

School Name: Amani Abeid Karume Secondary School

Diagnostic Scope: 90-Day & 215-Day Windows

Cache System: Squid Proxy Server

Overall Rating: EXCELLENT

1. Diagnostic Window Analytics

Analytical diagnostics captured across the 90-day and 215-day tracking windows indicate sustained terminal usage and high user concurrency. System logs show optimal distribution of user sessions throughout morning and afternoon laboratory hours.

Metric Window	Active Terminals	Avg Session Hours / Day	Squid Proxy Cache Hit Rate	System Uptime %
90-Day Diagnostic Block	42 / 42 Units	6.8 Hours	84.2%	99.4%
215-Day Diagnostic Block	42 / 42 Units	6.4 Hours	81.7%	98.9%

2. Network & Squid Proxy Efficiency

The local Squid Proxy Caching system operates at high efficiency, caching static educational content, media files, and offline library pages. This significantly reduces redundant bandwidth load and ensures instant page load speeds for thin client terminals during simultaneous student access.

3. Application & Resource Utilization Breakdown

eLearning Resource	Usage Frequency	Primary User Group	Resource Type
Khan Academy Lite	Daily High Volume	O-Level Students (Math/Science)	Offline Interactive Video & Exercises
Wikipedia Offline Engine	Daily High Volume	All Form Levels	General Academic Reference
Web Development Tools	4 Sessions / Week	Computer Studies Classes	Offline IDE & Web Frameworks

4. Technical Diagnostics & Recommendations

Squid proxy log recycling and cache partition trim operations should remain automated. Diagnostic metrics confirm zero hardware memory bottlenecks on the primary host server during peak 42-terminal concurrent booting.